

BS: Take the Hint - Interactive Multitracer PET/CT Lesion Segmentation with a Scribble-Conditioned ResEnc U-Net

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Abstract. Automated lesion segmentation in whole-body PET/CT is complicated by the variety of physiological tracer uptake patterns and by the differing appearance of lesions across tracers. The autoPET/CT V challenge addresses this by making segmentation *interactive*: user scribbles marking foreground and background are supplied alongside the image, and the algorithm is expected to exploit them. We present our submission, a scribble-conditioned residual encoder U-Net operating on four input channels – CT, PET, and a sparse scribble map for each of foreground and background. The network is initialised from the autoPET-III winning weights and extended from two to four input channels, with the two scribble channels zero-initialised so that the pretrained representation is preserved exactly at initialisation. Every model is fine-tuned per fold from the corresponding autoPET-III fold checkpoint, so that no validation case is seen during pretraining. PET intensities are normalised against a per-scan aorta blood-pool reference derived from a CT segmentation, which removes tracer- and centre-specific scaling without requiring lesion labels. At inference the five fold models are ensemble by averaging their softmax outputs per sliding-window patch, before Gaussian-weighted stitching. On the challenge’s five-fold split, with each fold evaluated on its own validation cases, mean Dice is 0.554 and mean lesion-level F1 is 0.528 without scribbles, rising to 0.751 and 0.733 after five correction rounds. About 85% of that gain follows the first scribble, and the spread between fold models narrows five-fold over the same rounds, so interaction largely compensates for how well or badly a given model segments unaided.

Keywords: autoPET challenge · Interactive segmentation · PET/CT

1 Introduction

Positron emission tomography combined with computed tomography (PET/CT) is central to oncological staging and therapy response assessment, and quantitative read-out of tumour burden requires lesion segmentation. Manual delineation of whole-body studies is slow and subject to inter-reader variation, which motivates automated methods.

The autoPET challenge series [11], built on a publicly released whole-body PET/CT cohort with manual lesion annotations [6–8], has driven progress on this task. Its fifth edition (autoPET/CT V) differs from the earlier editions in two respects. First, it is *multitracer*: the cohort contains both ^{18}F -FDG and $^{68}\text{Ga}/^{18}\text{F}$ -PSMA studies, whose physiological uptake patterns differ substantially, so that a model tuned to one tracer does not transfer to the other without care. Second, and centrally, the task is *interactive*: for each study the challenge supplies user scribbles that mark lesion (foreground) and non-lesion (background) voxels, and the algorithm is expected to use them. The clinically relevant question is therefore not only how well a model segments unaided, but how reliably it responds when a reader points at what it got wrong.

This paper describes our submitted algorithm. We treat the scribbles as two additional input channels to a three-dimensional residual encoder U-Net, initialise from the autoPET-III winning solution of Rokuss et al. [1], and fine-tune on the challenge data. We report five-fold cross-validation results over successive interactive correction rounds, which show that segmentation quality improves consistently as scribbles accumulate, most of the gain arriving with the first scribble.

2 Methods

2.1 Data

Training data We use exclusively the training data provided by the challenge: 1611 whole-body PET/CT studies [6–8], comprising 1014 ^{18}F -FDG and 597 PSMA examinations; the PSMA cohort combines 369 studies acquired with an ^{18}F -labelled and 228 with a ^{68}Ga -labelled PSMA ligand; we use the v2 data release, which corrects a small number of segmentation masks relative to the initial release. Each study is accompanied by a manual lesion annotation. No external or private data is used. For each study the challenge additionally provides foreground and background scribbles; these are used as supplied and are described in Section 2.2.

Validation data We follow the five-fold split distributed with the challenge data in `splits_final.json`, which divides the 1611 studies into folds of 1288/323 and 1289/322 training/validation cases. All results in Section 3 are obtained by evaluating each fold’s model on that same fold’s validation split, so that every one of the 1611 studies is predicted exactly once by a model that did not see it during training. The same split is used by the autoPET-III reference solution of Rokuss et al. [1], from which our weights are initialised, and we initialise each fold from that solution’s checkpoint for the same fold; consequently no validation case is seen during pretraining either. The per-fold checkpoints are those published by the authors via Zenodo [2].

2.2 Data pre-processing

All channels are resampled onto one common grid, defined from the original CT geometry: the voxel spacing is set to $2.0364 \times 2.0364 \times 3.0$ mm and the grid size to $\text{round}(\text{size} \cdot \text{spacing} / \text{target spacing})$ per axis, retaining the CT origin and direction. The identical procedure is applied at training time and inside the inference container.

CT. Intensities are clipped to $[-811.73, 1137.42]$ HU and z-scored with the dataset-wide statistics $\mu = 124.578$, $\sigma = 280.883$, then resampled with third-order B-spline interpolation and stored as `float16`.

PET. Standardised uptake values are divided by a per-scan reference and passed through an inverse hyperbolic sine before a fixed dataset-wide z-score with $\mu = 0.0904$, $\sigma = 0.2538$:

$$\text{PET}_{\text{norm}} = (\text{arcsinh}(\text{SUV}/(r + \varepsilon)) - \mu) / \sigma, \quad \varepsilon = 10^{-8}.$$

The reference r is the robust blood-pool activity of the aorta. An aortic mask is obtained by running TotalSegmentator [5] (`task="total"`, `fast=True`) on the CT, transferring the mask to the PET grid by nearest-neighbour interpolation and eroding it by one voxel to suppress partial-volume contamination from the vessel wall. Within the mask, focal outliers are removed at $\text{median} + 3 \cdot 1.4826 \cdot \text{MAD}$ and r is the mean of the remaining voxels. If the eroded mask contains fewer than 100 voxels with positive uptake, the reference falls back to the median of soft tissue ($-100 < \text{HU} < 100$), and failing that to the median over all positive PET voxels. This makes the normalisation label-free and tracer-agnostic. The normalised volume is resampled with B-spline interpolation and stored as `float16`.

Scribbles. The scribble heat maps are used exactly as provided by the challenge and are not regenerated. In the reference implementation they are produced by sampling up to $k = 5$ connected components of the annotation (26-connectivity). For each sampled component the single axial slice in which it occupies the most voxels is selected, and one scribble is drawn on that slice with the default centreline strategy, one of three strategies in the reference simulator, whose designs follow ScribblePrompt [9] and nnInteractive [10]: the two-dimensional slice mask is skeletonised, the largest skeleton component is retained (8-connectivity), the longest path between two skeleton endpoints is extracted and truncated by a fraction of 0.1. Background scribbles are drawn analogously inside a thin shell obtained by twice dilating the component with a unit-radius ball and subtracting the component itself – a two-voxel ring hugging the lesion boundary. At evaluation time background scribbles instead mark false-positive regions of the prediction, so the training and test semantics of the background channel differ; the interactive results in Section 3 measure the net effect of this mismatch.

Each scribble is therefore a planar curve lying in one axial slice, not a three-dimensional structure. The resulting scribble voxels are written into a volume on the PET grid with value 1 and convolved with a Gaussian of $\sigma = 0$, which is an identity operation; the maps are therefore *binary point maps* rather than smoothed heat maps, and are extremely sparse (of the order of 10^{-6} of the volume by occupancy). Foreground and background scribbles are stored as two separate volumes. We resample them to the target spacing with linear interpolation, clip to $[0, 1]$, and quantise to `uint8` as $\lfloor 255x \rfloor$; they are rescaled to $[0, 1]$ by division by 255 when a patch is loaded.

Annotations. Lesion masks are resampled to the target spacing by nearest-neighbour interpolation.

2.3 Algorithm/model

The network is a three-dimensional residual encoder U-Net following the nnU-Net [4] ResEncL preset, configured with four input channels, two output classes and deep supervision enabled. The four channels are CT, normalised PET, the foreground scribble map and the background scribble map, in that order.

Weights are initialised from the autoPET-III winning solution of Rokuss et al. [1], which is itself pretrained on a large multi-dataset corpus (`Dataset619_nativemultistem`) and then fine-tuned on the autoPET-III cohort. Two adaptations are made. First, the input stem is extended from the two channels of the original model (CT and PET) to four; the convolution weights of the two additional scribble channels are *zero-initialised*, so that at initialisation the extended network computes exactly the function of the pretrained two-channel model and the scribble pathway is learned from scratch during fine-tuning. Second, the pretrained model carries an auxiliary organ segmentation head, which we disable during fine-tuning: the organ information is assumed to be retained in the encoder representation and is not supervised further. Fold i of our model is initialised from fold i of the pretrained model, so the fold structure is preserved end to end.

2.4 Data post-processing

Inference is performed by sliding a $192 \times 192 \times 192$ window over the resampled volume with an overlap of 0.5, i.e. a stride of 96 voxels along each axis, with the last window on each axis shifted inward so that it fits inside the volume. Should an axis be shorter than the patch, the volume is zero-padded at its far end and the result cropped back afterwards. The forward passes run in mixed precision, as during training.

The per-patch class posteriors, obtained by a softmax over the network logits, are accumulated into the output volume with a Gaussian importance weight, which down-weights the poorly contextualised patch borders relative to the patch centre. The weight map is obtained by placing a unit impulse at the patch centre, convolving it with a Gaussian whose standard deviation is $\sigma_{\text{scale}} = 0.125$ of the

patch size along each axis (i.e. 24 voxels for a patch of 192), normalising by its maximum and clipping from below at 10^{-3} . Two accumulators are maintained: the sum of weighted class probabilities and the sum of the weights themselves; their element-wise ratio yields the final probability volume, which is correct irrespective of how many windows cover a given voxel.

The foreground channel of this probability volume is then resampled back to the original acquisition grid by linear interpolation and clipped to $[0, 1]$, and only afterwards thresholded at 0.65 to produce the binary lesion mask, which is written on the reference volume’s grid and affine. Applying the threshold after rather than before resampling avoids nearest-neighbour interpolation of a binary mask, which would otherwise discard small lesions. The threshold was selected by a sweep on fold 0’s validation split and applied unchanged to all folds and to the ensemble. No connected-component filtering, size thresholding or other morphological post-processing is applied.

2.5 Training and test parameters

Each fold is trained for 500 epochs with stochastic gradient descent at a learning rate of 10^{-3} , Nesterov momentum of 0.99, zero dampening and a weight decay of 3×10^{-5} . No learning rate schedule is used. The batch size is 2, training runs in mixed precision, and gradients are clipped to a maximum norm of 12.

The loss is the sum of cross-entropy and a soft Tversky term,

$$\mathcal{L} = \text{CE} + \left(1 - \frac{1}{|C|} \sum_{c \in C} \frac{\text{TP}_c + \epsilon}{\text{TP}_c + \alpha \text{FP}_c + \beta \text{FN}_c + \epsilon}\right),$$

with $\alpha = 0.3$, $\beta = 0.7$ and $\epsilon = 10^{-5}$. The Tversky term is averaged over foreground classes only; including the background would cancel the asymmetry between α and β , since a foreground false negative is a background false positive. Penalising false negatives more heavily than false positives reflects the clinical cost of missing a lesion. The background remains supervised through the cross-entropy term. Deep supervision is applied with the standard nnU-Net weighting $[1, \frac{1}{2}, \frac{1}{4}, \dots]$, normalised to unit sum and with the coarsest resolution level dropped.

Training patches are sampled stochastically rather than augmented. For a tumour-bearing case, with probability 0.33 a hard-negative, tumour-free window is drawn (the first such window found within 20 attempts, otherwise the window containing the fewest lesion voxels); otherwise a lesion is drawn *uniformly at the lesion level*, independently of its size, and a window is placed at random such that it contains a random voxel of that lesion. Tumour-free cases receive a uniformly random window. No intensity or geometric augmentation of any kind is applied.

For every fold the checkpoint with the lowest validation loss over the 500 epochs is selected, using the same criterion for all five folds. Training was performed on a single NVIDIA V100 with 32 GB of memory. At test time no test-time augmentation is used. The five fold models are ensembled by averaging their softmax outputs *per sliding-window patch*, before the patch is stitched into the

output volume, so that the Gaussian weighting and reverse resampling described above operate on a single already-ensembled probability map.

2.6 Github repository

Link to Github repository: https://github.com/amged-elmasry-bs/bs_autopetV_submission

3 Results

We report five-fold cross-validation on the challenge training data using the `splits_final.json` split, evaluating each fold’s model on that fold’s validation cases. All scores are produced by the challenge’s official `MetricEvaluator` [3], using its two metrics unmodified.

The first is the volumetric Dice between the predicted mask P and the annotation G ,

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|}.$$

It is evaluated on the original acquisition grid. When the annotation contains no lesion, $|G| = 0$, the quotient is undefined and the evaluator returns `NaN`. This depends on the annotation alone: a lesion-free study yields `NaN` whether the model correctly predicts nothing or produces a false positive, so such cases carry no Dice at all rather than a score of zero or one.

The second is a lesion-level F_1 . Both volumes are decomposed into connected components at 18-connectivity, and a predicted component is counted as a true positive when its intersection over union with an annotated lesion is at least 0.1; unmatched predicted components are false positives and unmatched annotated lesions are false negatives. With those *lesion counts*,

$$F_1 = \frac{2 \text{ TP}}{2 \text{ TP} + \text{FP} + \text{FN}}.$$

Every lesion contributes equally regardless of its size, and the 0.1 overlap requirement means the metric asks whether a lesion was detected rather than how precisely it was delineated. F_1 is likewise undefined when the annotation contains no lesion, since $\text{TP} + \text{FN} = 0$.

Studies without annotated lesions are therefore excluded from both averages, leaving 1040 of the 1611 studies scored, between 199 and 215 per fold. Two further conventions matter for comparability. First, alongside the per-case macro average reported here, the evaluator pools lesion counts across *all* cases into an aggregated F_1 , in which false positives on tumour-free studies do count. Second, the challenge ranks by the area under each metric’s six-round curve, a trapezoid over the round index in which the endpoint rounds r_0 and r_5 carry half the weight of the interior ones. We report it normalised by the number of intervals, so that it lies on the same $[0, 1]$ scale as the metric it integrates; per-fold AUCs

are given in Tables 1 and 2. The reported figures follow the same convention as the official challenge leaderboard, which reports a count of scored cases smaller than the number submitted for exactly this reason. Note that these are single-model results: each fold model is scored only on the cases it did not train on. The five-fold ensemble that constitutes the actual submission cannot be evaluated in this protocol, since for any training case four of its five members have seen that case; its performance is measured on the held-out challenge test set.

Performance is reported over successive interactive rounds. Round r_0 is the unaided prediction, with both scribble channels empty. In each subsequent round one scribble is generated from the model’s current errors and appended to the accumulated input, following the challenge’s reference interactive loop: a candidate scribble is drawn on the false-positive region ($\hat{y} = 1, y = 0$) and another on the false-negative region ($\hat{y} = 0, y = 1$), both by the challenge’s own scribble generator. For each error region it examines every axial slice, takes the largest two-dimensional connected component (8-connectivity) within each, and selects the slice whose component is largest overall; the scribble is the centreline of that component and lies entirely in that one slice. Of the two candidates the *longer* is kept – a foreground scribble when the false-negative scribble is at least as long as the false-positive one, and a background scribble otherwise. A round therefore adds a single planar curve, typically a few voxels in extent, to whichever error the generator could trace furthest. Scribbles are encoded exactly as at training time and as in the submitted container: written as ones on the original grid, re-sampled linearly to the target grid, clipped and quantised through `uint8`, with no additional smoothing or renormalisation.

Each case therefore receives **six inference passes in total**: r_0 without any scribble, followed by five refinement rounds $r_1 \dots r_5$, each adding one further scribble to those already accumulated.

Table 1. Five-fold cross-validation, Dice over the six interactive rounds (r_0 without scribbles, then five refinement rounds). Each fold’s model is evaluated on its own validation split. Both the per-round means and the AUC are computed over tumour-bearing cases only, since Dice is undefined on a lesion-free annotation and the evaluator returns `NaN`. The AUC is the trapezoid of the six-round curve over the round index, normalised by the number of intervals so that it shares the scale of the metric itself and its maximum is 1.0. The Mean row averages the five fold curves.

Fold	r_0	r_1	r_2	r_3	r_4	r_5	AUC
0	0.5708	0.7344	0.7536	0.7591	0.7637	0.7654	0.7358
1	0.4583	0.7130	0.7424	0.7523	0.7598	0.7638	0.7157
2	0.5374	0.7164	0.7242	0.7281	0.7300	0.7310	0.7066
3	0.6312	0.7291	0.7422	0.7480	0.7528	0.7540	0.7330
4	0.5719	0.7186	0.7328	0.7390	0.7414	0.7416	0.7177
Mean	0.5539	0.7223	0.7391	0.7453	0.7495	0.7512	0.7218

Table 2. Five-fold cross-validation, lesion-level F_1 over the six interactive rounds. As in Table 1, the means and the AUC are computed over tumour-bearing cases only, F_1 being undefined when the annotation contains no lesion, and the AUC is the normalised trapezoid over the round index (maximum 1.0).

Fold	r_0	r_1	r_2	r_3	r_4	r_5	AUC
0	0.5327	0.7160	0.7359	0.7388	0.7493	0.7506	0.7163
1	0.4394	0.6908	0.7139	0.7232	0.7302	0.7395	0.6895
2	0.5103	0.7010	0.7066	0.7093	0.7122	0.7135	0.6882
3	0.6142	0.7062	0.7250	0.7272	0.7334	0.7347	0.7133
4	0.5437	0.7039	0.7178	0.7230	0.7224	0.7246	0.7002
Mean	0.5281	0.7036	0.7198	0.7243	0.7295	0.7326	0.7015

Averaged over the five folds, the unaided model reaches a Dice of 0.5539 and a lesion-level F_1 of 0.5281. A single scribble raises these to 0.7223 and 0.7036, and after five rounds they stand at 0.7512 and 0.7326 – a total gain of +0.1973 Dice and +0.2045 F_1 . The gain is heavily front-loaded: the first scribble alone accounts for 85% of it in both metrics, with the remaining four rounds contributing +0.0289 Dice and +0.0290 F_1 between them. The curves are therefore steep at r_1 and close to flat by r_3 , which is what the normalised AUC of 0.7218 (Dice) and 0.7015 (F_1) reflects.

Dice increases at every round on all five folds. F_1 does so on four of them; fold 4 dips by 0.0006 between r_3 and r_4 before recovering, an excursion far smaller than the between-fold spread and the only non-monotonic step in the sixty transitions recorded. At the level of individual studies, 497 of the 1040 scored cases improve in F_1 from r_0 to r_5 , 461 are unchanged and 82 deteriorate.

The most pronounced effect is on the spread between folds. Unaided, the folds differ by 0.1729 Dice (0.4583 on fold 1 to 0.6312 on fold 3); after five rounds that spread is 0.0344 (0.7310 to 0.7654), a five-fold compression, with the corresponding F_1 spread falling from 0.1748 to 0.0371. The ordering is not preserved either: fold 1 is the weakest fold unaided and the second strongest at r_5 , whereas fold 2 – whose selected checkpoint also has the highest validation loss of the five – gains least after the first scribble and ends last. Interaction thus compensates for much of the variation in unaided quality between fold models, which is the behaviour one wants from a corrective interface.

4 Discussion

The cross-validation results in Tables 1 and 2 show that segmentation quality increases with each interactive round: the model follows the supplied scribbles and progressively corrects its own errors rather than ignoring the additional channels or degrading as more input is accumulated.

Averaged over folds the loop is worth +0.1973 Dice and +0.2045 F_1 , and the first scribble supplies roughly 85% of that; the four subsequent rounds refine rather than transform the segmentation.

Where the gain comes from is worth separating out. Pooled over all folds and cases, five rounds convert 829 false negatives into true positives, while the false positive count rises slightly, from 6869 to 6998. The improvement is thus almost entirely a matter of recovering lesions the model had missed, not of suppressing spurious ones. This is despite the loop spending most of its budget on the latter: under the challenge’s rule the longer of the two candidate scribbles is kept, and because false-positive regions are typically larger and more elongated than the residual false-negative ones, 69% of the 5426 scribbles drawn are background scribbles. Those background scribbles are, on this evidence, close to inert. The likely reason is the semantic mismatch noted in Section 2.2: during training the background channel carries a thin shell hugging a true lesion, marking its boundary, whereas at evaluation it marks a whole false-positive region, and the network has never been trained to read it that way. Training with evaluation-style background scribbles – drawn on the model’s own false positives rather than around annotated lesions – is the most promising single change suggested by these results.

Two design choices are worth noting. Normalising PET against a per-scan aorta blood-pool reference is label-free and does not depend on the tracer, which matters for a cohort combining FDG and PSMA studies whose absolute uptake scales differ. Zero-initialising the two scribble channels means the extended four-channel network reproduces the pretrained two-channel model exactly before any fine-tuning, so the interactive capability is added without disturbing the segmentation behaviour inherited from the autoPET-III solution.

Several limitations should be stated. The cross-validation figures are single-model results; the submitted five-fold ensemble cannot be scored under this protocol, so its benefit is assumed from the usual behaviour of nnU-Net ensembles rather than measured here. The threshold of 0.65 was chosen on fold 0 and reused unchanged everywhere, including for the ensemble, whose probability distribution is not identical to a single model’s. Scores are reported on tumour-bearing studies only, because both metrics are undefined without an annotated lesion; this excludes 571 of the 1611 studies and means that false positives on lesion-free scans, clinically the more costly error, do not enter the per-case averages at all. The simulated reader is also an optimistic one: scribbles are generated automatically from the ground truth by the challenge’s own simulator, always mark a genuine error, and are drawn as clean centrelines, so the results bound what a perfectly informed annotator would obtain rather than what a human would. Finally, the background channel behaves as described above, which caps what further rounds can achieve.

5 Conclusion

We described our submission to the autoPET/CT V challenge: a scribble-conditioned three-dimensional residual encoder U-Net, initialised per-fold from the autoPET-III winning weights, extended from two to four input channels with zero-initialised scribble pathways, trained with a false-negative-weighted Tversky and cross-entropy objective, and deployed as a five-fold ensemble whose fold predictions are averaged per sliding-window patch. Five-fold cross-validation shows that performance improves consistently across successive interactive rounds, confirming that the model uses the supplied scribbles to correct its own errors. Averaged over the five folds, Dice rises from 0.5539 unaided to 0.7512 after five scribbles and lesion-level F_1 from 0.5281 to 0.7326, with about 85% of the gain delivered by the first scribble, and the spread between fold models narrows five-fold over the same rounds.

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Table 3. Algorithm details

Team name	algorithm name	data pre-processing	data post-processing	training data augmentation
BS	BS baseline	Resampled to a common grid derived from the CT geometry at 2.0364 x 2.0364 x 3.0 mm. CT: clip to $[-811.7, 1137.4]$ HU, dataset z-score, B-spline. PET: arcsinh(SUV / per-scan aorta blood-pool reference), dataset z-score, B-spline; reference from a TotalSegmentator aorta mask with MAD outlier rejection. Scribbles: linear resample, clip $[0, 1]$, <code>uint8</code> .	Sliding window 192^3 at 0.5 overlap; per-patch softmax stitched with Gaussian importance weights ($\sigma = 0.125 \times$ patch); reverse linear resample to the original grid; threshold 0.65. No connected-component or size filtering.	None. Stochastic patch sampling only ($p_{\text{bg}} = 0.33$ hard negatives, otherwise uniform)
test time ensembling augmentation		standardized framework	network architecture	loss
None	5-fold cross-validation ensemble; softmax averaged per sliding-window patch, before Gaussian stitching	nnU-Net, ResEncL preset (3D)	Residual Encoder U-Net (3D); 4 input channels (CT, PET, foreground and background scribbles), 2 output classes, deep supervision	Cross-entropy + Tversky ($\alpha = 0.3$, $\beta = 0.7$, foreground classes only), deep-supervision weighted
training data	data/model dimensionality and size	use of pre-trained models	GPU hardware for training	
1014 FDG + 597 PSMA PET/CT of autoPET (369 ^{18}F - and 228 ^{68}Ga -labelled PSMA); 5-fold split of <code>splits_final.json</code>	3D; patch $192 \times 192 \times 192$ at 2.0364 x 2.0364 x 3.0 mm	Public: autoPET-III winning model of Rokuss et al. [1, 2], itself pretrained on <code>Dataset619_nativemultistem</code> . Fold i initialised from their fold i ; input stem extended $2 \rightarrow 4$ channels with the two scribble channels zero-initialised; auxiliary organ head disabled	1x NVIDIA V100 (32 GB)	